

ULYANOVSK, Russian Federation

WMO: 277850

Lat: 54.276N

Lon: 48.243E

Elev: 463

StdP: 14.45

Time Zone: 4.00 (E04)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.0	-13.9	-25.2	1.4	-19.7	-18.7	2.0	-13.5	27.2	28.6	24.7	24.0	5.0	340	0.627

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.8	89.2	67.9	85.6	67.5	82.2	65.8	70.9	82.4	69.3	81.1	67.7	78.9	9.3	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.9	100.8	76.2	65.2	94.9	74.6	63.6	89.5	73.2	35.0	82.3	33.7	81.2	32.4	79.0	78.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
24.6	22.4	20.2	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-25.7	93.0	7.3	3.8	-30.9	95.7	-35.1	97.9	-39.2	100.1	-44.5	102.8		
			-25.7	73.6	7.2	2.1	-30.8	75.2	-35.0	76.4	-39.0	77.6	-44.2	79.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	41.1	12.9	13.0	24.7	42.9	56.4	64.4	68.6	65.0	54.6	41.9	28.7	17.7	(6)
(7)		DBStd	22.34	13.34	13.62	9.88	9.78	8.79	7.34	6.29	7.24	7.97	8.06	9.87	12.66	(7)
(8)		HDD50	5221	1151	1036	785	255	40	2	0	0	40	273	639	1000	(8)
(9)		HDD65	9095	1616	1456	1250	662	290	98	36	93	323	717	1089	1465	(9)
(10)		CDD50	1956	0	0	0	44	237	435	576	464	179	21	0	0	(10)
(11)		CDD65	355	0	0	0	0	22	81	147	92	13	0	0	0	(11)
(12)		CDH74	3737	0	0	0	13	310	851	1485	961	117	0	0	0	(12)
(13)		CDH80	1214	0	0	0	0	73	259	502	359	21	0	0	0	(13)
(14)	Wind	WSAvg	9.0	9.3	9.1	9.4	9.6	9.4	8.4	7.4	7.7	8.4	9.6	10.0	9.4	(14)
(15)	Precipitation	PrecAvg	19.5	1.3	1.1	1.0	1.3	1.8	2.7	2.0	2.2	1.6	1.4	1.3		(15)
(16)		PrecMax	27.4	3.1	2.1	2.4	3.7	4.3	4.9	5.8	4.1	6.0	3.0	2.5	3.1	(16)
(17)		PrecMin	13.3	0.5	0.4	0.2	0.1	0.1	0.0	0.2	0.6	0.4	0.7	0.4	0.4	(17)
(18)		PrecStd	3.5	0.6	0.5	0.5	0.8	1.1	1.4	1.3	1.0	1.5	0.7	0.6	0.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.0	37.2	50.7	75.6	86.2	92.1	93.6	94.8	83.0	67.6	52.0	39.5	(19)
(20)			MCWB	35.3	34.6	43.0	56.2	64.1	67.2	67.1	67.8	62.5	53.8	48.5	38.1	(20)
(21)		2%	DB	34.5	35.5	42.9	70.2	81.6	86.6	88.9	88.0	77.1	62.1	47.3	36.6	(21)
(22)			MCWB	33.2	33.6	37.9	53.4	61.7	67.3	68.3	67.5	60.6	52.1	44.8	35.1	(22)
(23)		5%	DB	32.9	33.9	39.1	64.9	77.2	82.4	85.6	83.9	73.0	57.5	43.5	34.9	(23)
(24)			MCWB	31.7	32.4	35.2	51.5	59.9	66.0	68.6	66.8	59.3	51.1	41.5	33.5	(24)
(25)		10%	DB	30.6	31.6	36.5	59.7	72.8	78.8	82.1	78.9	68.1	53.7	40.6	33.3	(25)
(26)			MCWB	29.3	30.1	33.9	48.7	57.6	64.8	67.1	65.1	57.8	48.4	38.9	32.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.4	35.5	43.9	58.4	67.5	71.6	73.6	71.9	65.7	56.7	49.4	38.8	(27)
(28)			MCDWB	36.1	36.6	50.6	72.2	83.2	83.5	84.8	83.7	77.4	62.4	51.6	39.2	(28)
(29)		2%	WB	33.3	33.9	38.1	55.1	63.7	69.7	71.5	69.7	62.7	53.6	45.2	35.2	(29)
(30)			MCDWB	34.2	35.1	41.8	66.9	77.0	81.7	82.3	82.6	72.8	59.7	47.0	36.1	(30)
(31)		5%	WB	31.8	32.4	35.8	52.2	61.4	67.9	70.0	67.9	60.6	51.1	41.6	33.6	(31)
(32)			MCDWB	32.9	33.8	38.3	63.2	73.9	78.9	81.2	80.5	69.4	56.5	43.2	34.6	(32)
(33)		10%	WB	29.4	30.0	34.1	49.3	58.9	66.1	68.4	66.1	58.6	49.2	38.8	32.1	(33)
(34)			MCDWB	30.6	31.6	36.1	59.2	70.2	76.3	79.1	77.6	66.9	53.4	40.2	33.1	(34)
(35)	Mean Daily Temperature Range		MDBR	11.6	13.9	14.8	17.9	20.7	19.8	19.8	19.7	17.3	13.2	8.9	10.2	(35)
(36)		5% DB	MCDBR	8.5	9.3	14.8	26.0	27.4	25.6	25.4	27.2	24.3	20.1	11.1	7.4	(36)
(37)			MCWBR	8.5	8.4	10.4	13.6	12.6	11.3	10.2	11.8	11.8	12.6	9.4	7.1	(37)
(38)			MCDBR	8.4	9.0	12.8	24.0	24.2	21.9	21.7	23.5	21.4	17.8	10.0	7.1	(38)
(39)		5% WB	MCWBR	8.5	8.4	9.4	13.3	12.1	10.9	10.0	11.5	11.6	12.2	9.0	7.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.287	0.319	0.359	0.417	0.393	0.385	0.400	0.425	0.375	0.350	0.316	0.287	(40)	
(41)		taud	2.092	2.060	2.044	2.124	2.320	2.382	2.339	2.253	2.372	2.346	2.236	2.087	(41)	
(42)		Ebn at Noon	200	235	252	255	271	274	266	251	250	227	191	171	(42)	
(43)		Edh at Noon	22	32	41	43	37	36	37	38	30	24	19	18	(43)	
(44)	All-Sky Solar Radiation	RadAvg	253	568	1014	1342	1761	1883	1868	1493	950	466	197	142	(44)	
(45)		RadStd	38	69	117	124	129	176	161	132	131	80	33	37	(45)	

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	N/A	+2.30	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+203	N/A	(47)

Nomenclature: See separate page